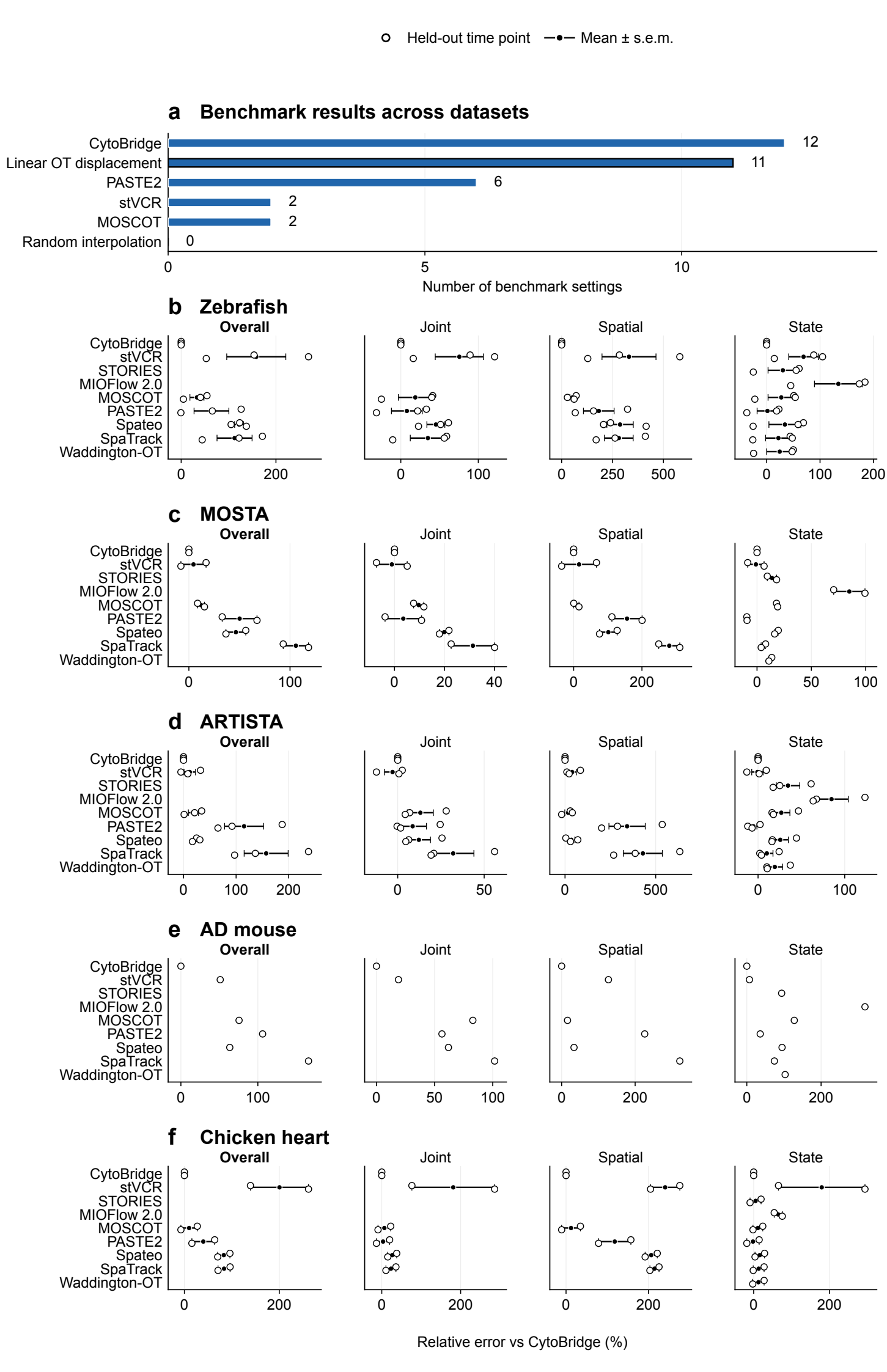

Held-out time point
Mean ± s.e.m.
a Benchmark results across datasets
CytoBridge
Linear OT displacement
PASTE2
stVCR
MOSCOT
Random interpolation
12
11
6
2
2
0
0
5
10
Number of benchmark settings
b Zebrafish
c MOSTA
d ARTISTA
e AD mouse
f Chicken heart
Overall
Joint
Spatial
State
CytoBridge
stVCR
STORIES
MIOFlow 2.0
MOSCOT
PASTE2
Spateo
SpaTrack
Waddington-OT
Relative error vs CytoBridge (%)